MINISTERUL EDUCAȚIEI
UNIVERSITATEA TEHNICĂ
DIN CLUJ-NAPOCA

PROIECT

PROIECTARE INFRASTRUCTURĂ TRAMVAI-METROU

Proiectarea unui traseu de linie de tramvai

plan, profil longitudinal, profile transversale

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Secția IUDR
Grupa 3641

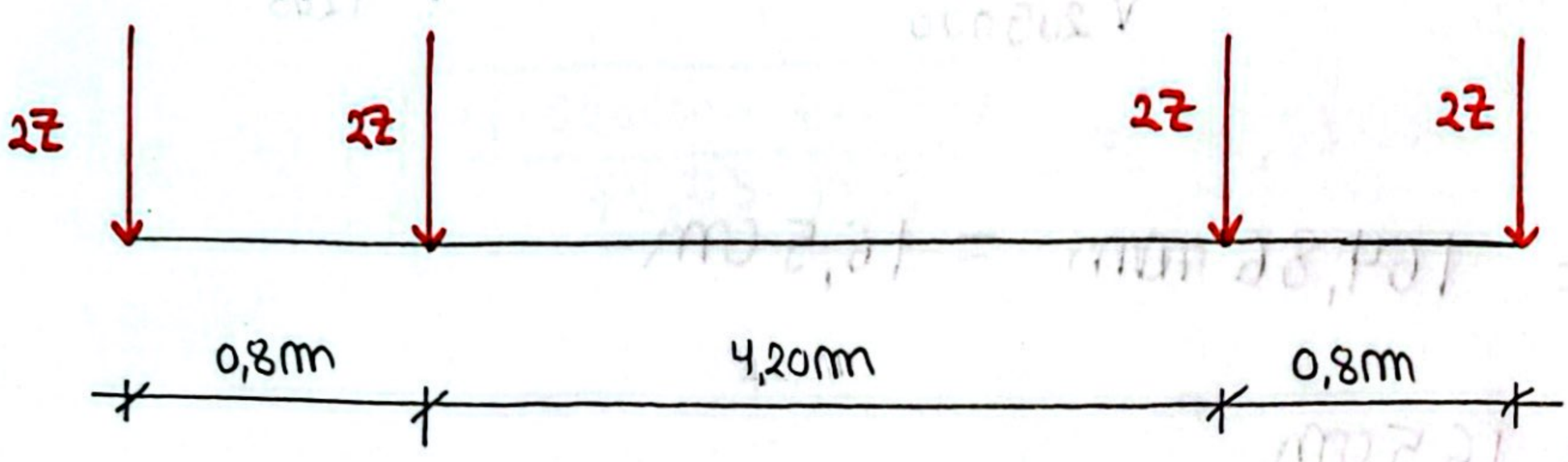
Cluj Napoca, 2024

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HANAPCIUC
BERNARD
ALEXANDRU
IUDR3641
2024-2025

Dimens. suprastructurii rigide - TRAMVAI

CONVOI 1 $\quad n = 10$



$2Z = 65,85 kN$

$h_1 = (25 + 0,1 \cdot 10)\ cm = 26 cm$

$h_2 = (30 + 0,1 \cdot 10)\ cm = 31 cm$

$E_1 = 220\,000 \ \frac{daN}{cm^2}$

$E_2 = 205\,000 \ \frac{daN}{cm^2}$

$E_3 = 1200 \ \frac{daN}{cm^2}$

h_1 26cm	$E_1 = 220\,000$
h_2 31cm	$E_2 = 205\,000$
h_3 16,5cm	$E_3 = 1200$

1) Coef. de pat efectiv

$$\underline{h}^x = 0{,}83 \cdot h_1 \cdot \sqrt[3]{\frac{E_1}{E_2}} + C \cdot h_2 \cdot \sqrt[3]{\frac{E_2}{E_3}}$$

$$\underline{h}^x = 0{,}83 \cdot 26 \cdot \sqrt[3]{\frac{220000}{205000}} + 0{,}83 \cdot 31 \cdot \sqrt[3]{\frac{205000}{1200}}$$

22,94

$$\underline{h}^x = 164{,}86 \text{ mm} = 16{,}5 \text{ cm}$$

$$\underline{h}^x = 16{,}5 \text{ cm}$$

$$\underline{k} = \frac{E_3}{h^x} = \frac{1200 \frac{daN}{cm^2}}{16{,}5 \text{ cm}} = 72{,}73$$

2) Grosime echivalentă bistrat

$$\underline{h_e} = \sqrt[3]{\frac{E_1 \cdot h_1^3 + E_2 \cdot h_2^3}{E_1}} = \sqrt[3]{\frac{220000 \cdot 26^3 + 205000 \cdot 31^3}{220000}}$$

$$\underline{h_e} = 35{,}66 \text{ cm}$$

3) Mom. încovoietor

$$\underline{M_e} = \frac{L_e}{4} \cdot \Sigma f_i \mu_i$$

$$\underline{L_e} = \sqrt[4]{\frac{4 \cdot E_e \cdot I_e}{k \cdot b_e}}$$

$$I_e = \frac{b_e h_e^3}{12} = \frac{17{,}83 \cdot 35{,}66^3}{12} = 67377{,}38 \text{ cm}^4$$

$$b_e = \frac{h_e}{2} = \frac{35{,}66}{2} = 17{,}83 \text{ cm}$$

$$L_e = \sqrt[4]{\frac{4 \cdot 220000 \cdot 67377{,}38}{72{,}73 \cdot 17{,}83}} = 82{,}23 \text{ cm}$$

$x_1 = 0 \Rightarrow \mu(x1) = 1$

$x_2 = 0{,}8m \Rightarrow \mu(0{,}8) = e^{-\frac{10{,}81}{1{,}12}} \left(\cos\frac{10{,}81}{1{,}12} - \sin\frac{10{,}81}{1{,}12}\right) = 0{,}108$

$x_3 = 5m \Rightarrow \mu(5) = e^{-\frac{151}{1{,}12}} \left(\cos\frac{151}{1{,}12} - \sin\frac{151}{1{,}12}\right) = 0{,}0236$

$x_4 = 5{,}8m \Rightarrow \mu(5{,}8) = e^{\frac{-15{,}81}{1{,}12}} \left(\cos\frac{15{,}81}{1{,}12} - \sin\frac{15{,}81}{1{,}12}\right) = 0{,}0197$

$\Sigma \mu(x) = 1{,}15 m = 115 cm$

$$M_e = \frac{65{,}85 \cdot 10^2 \cdot 82{,}23}{4} \cdot 115 = 1556{,}77 \text{ kNm}$$

4) Mom. încov. în strat 1,2

$$M_1 = M_e \cdot \frac{E_1 h_1^3}{E_1 h_1^3 + E_2 h_2^3} = 1556{,}77 \cdot 10^4 \cdot \frac{220000 \cdot 26^3}{220000 \cdot 26^3 + 205000 \cdot 31^3}$$

$$M_1 = 603{,}54 \text{ kNm}$$

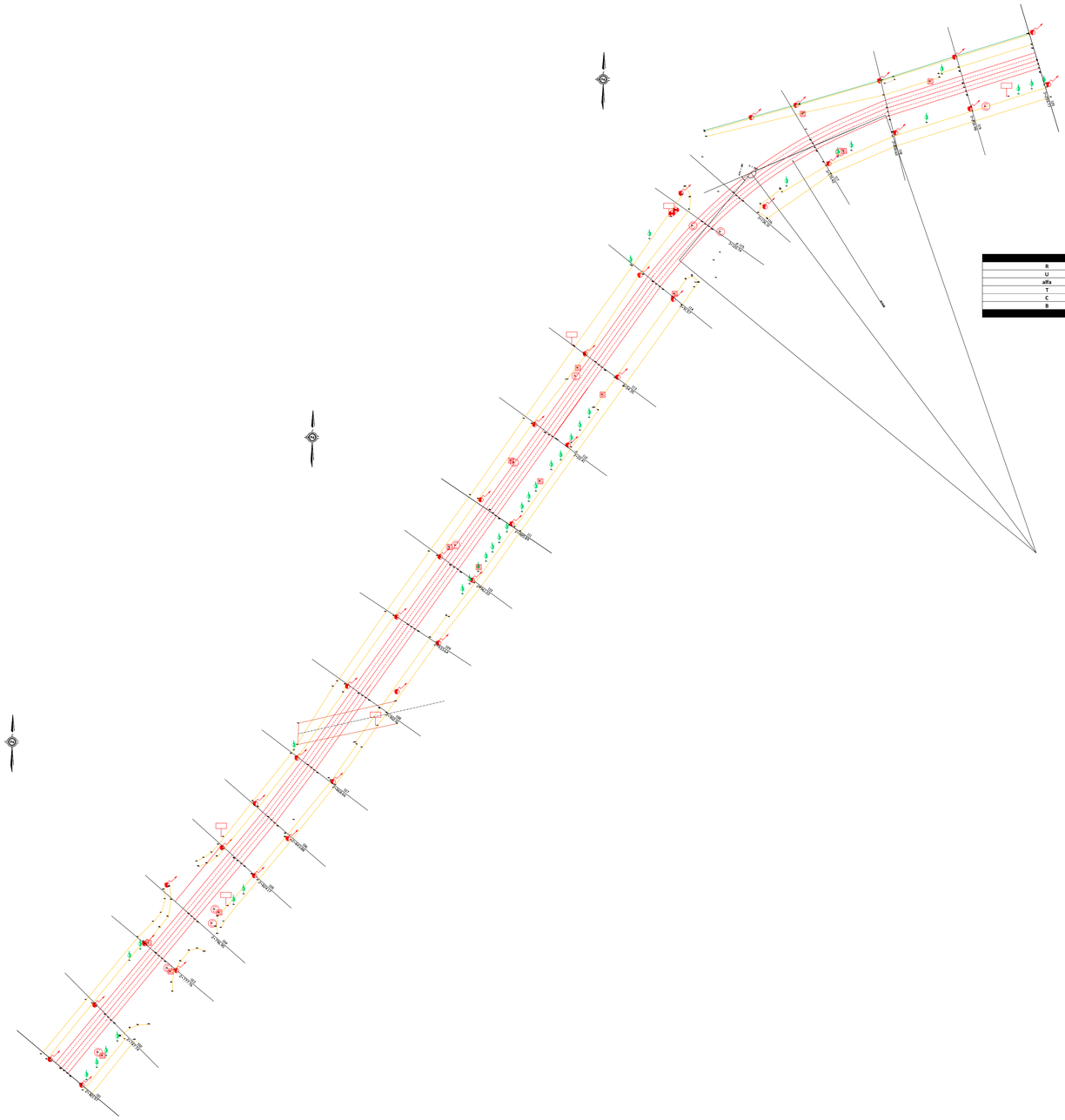
$$M_2 = M_e \cdot \frac{E_2 h_2^3}{E_1 h_1^3 + E_2 h_2^3} = 1556{,}77 \cdot 10^4 \cdot \frac{205000 \cdot 31^3}{220000 \cdot 26^3 + 205000 \cdot 31^3}$$

$$M_2 = 953{,}23 \text{ kNm}$$

5) Ef. unitare normale în strat 1,2

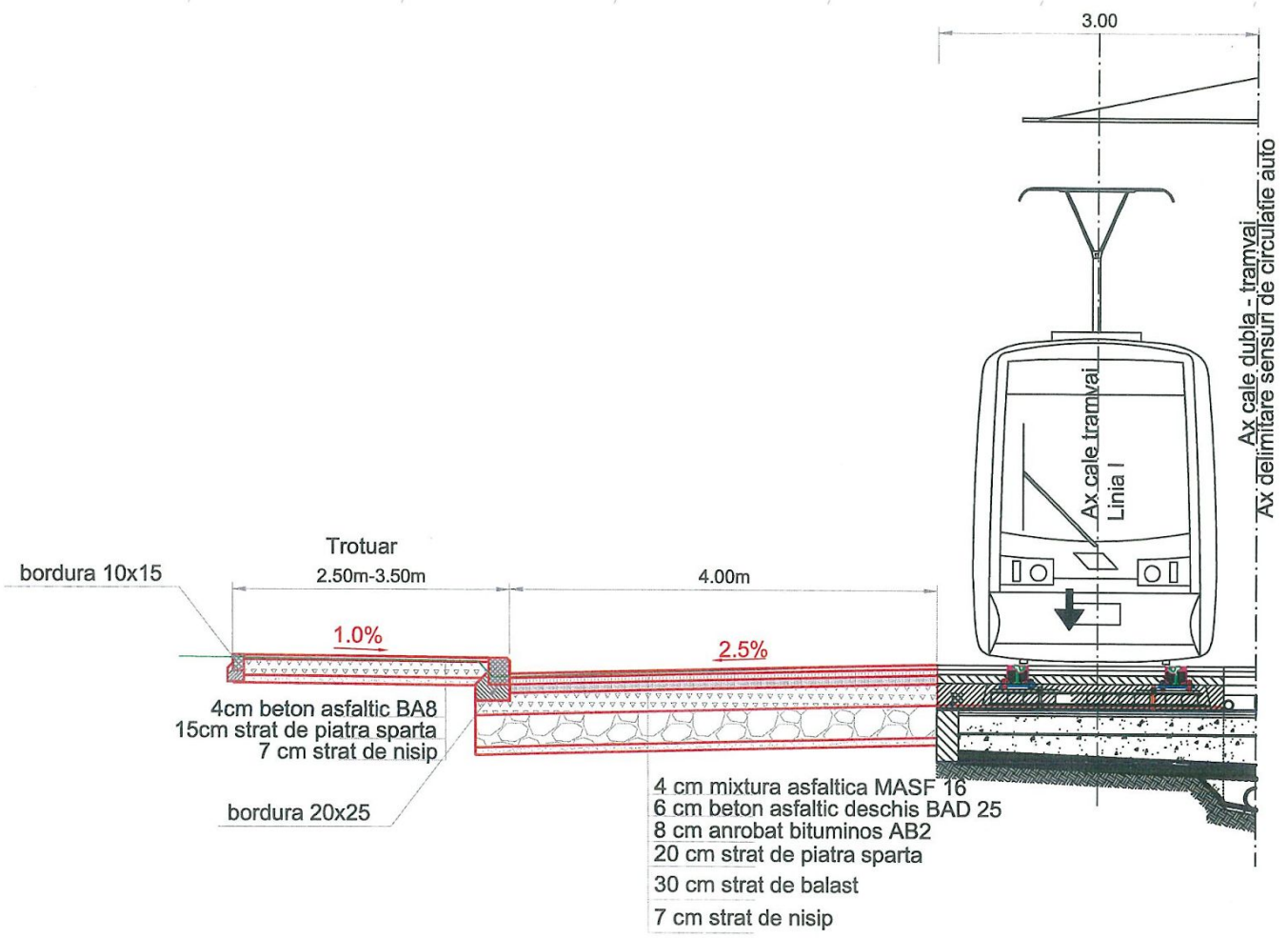
$$\sigma_{y1} = 6 \frac{M_1}{h_1^2} = 6 \cdot \frac{603{,}54}{26^2} = 5{,}36 \frac{kN}{cm^2} = 53{,}6 \text{ MPa}$$

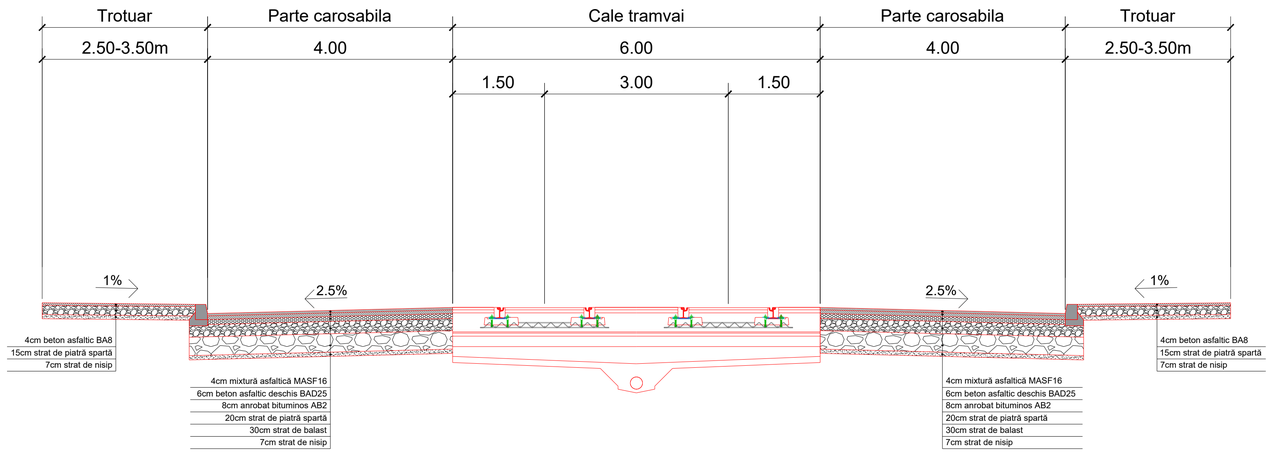
$$\sigma_{y2} = 6 \frac{M_2}{h_2^2} = 6 \cdot \frac{953{,}23}{31^2} = 5{,}95 \frac{kN}{cm^2} = 59{,}5 \text{ MPa}$$


R170
R170
R170

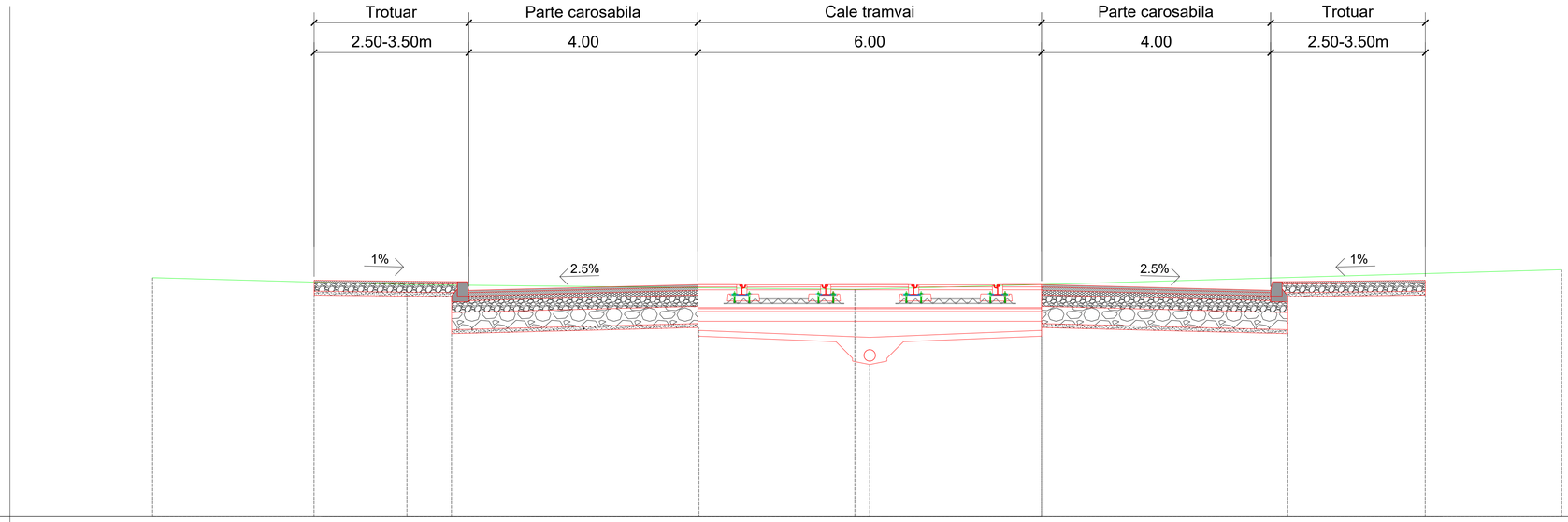

z
1:100
345.00
340.00
335.00

| | | |

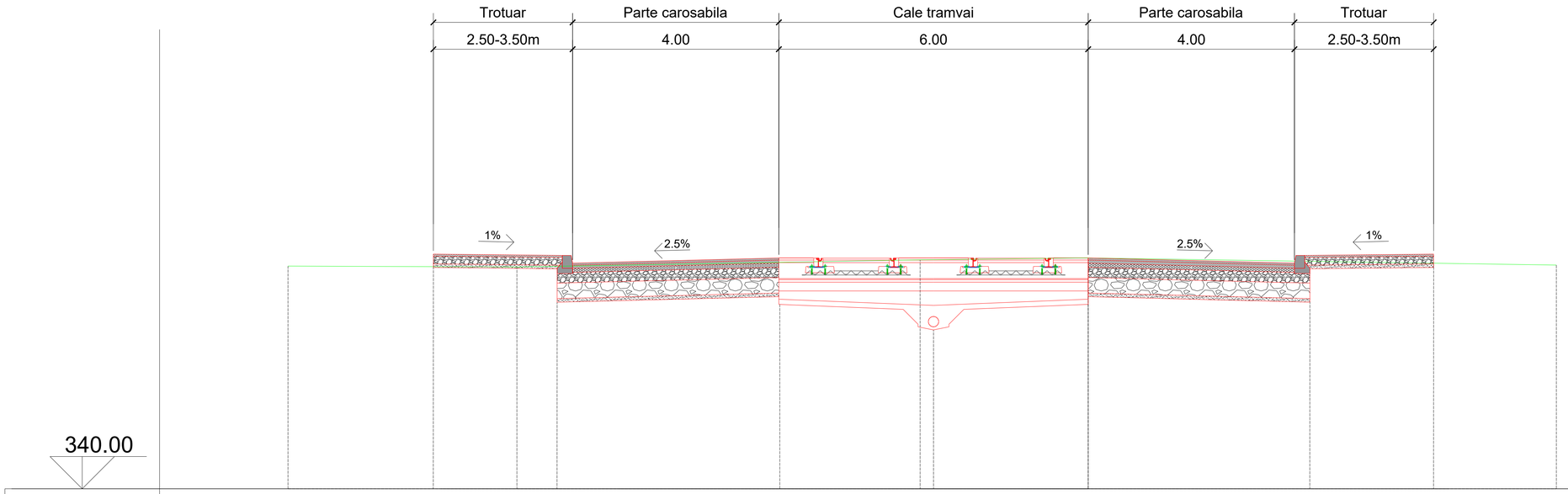

3.00
Ax cale dubla - tramvai
Ax delimitare sensuri de circulatie auto
Ax cale tramvai
Linia I
Trotuar
bordura 10x15
2.50m-3.50m
4.00m
1.0%
2.5%
4cm beton asfaltic BA8
15cm strat de piatra sparta
7 cm strat de nisip
bordura 20x25
4 cm mixtura asfaltica MASF 16
6 cm beton asfaltic deschis BAD 25
8 cm anrobat bituminos AB2
20 cm strat de piatra sparta
30 cm strat de balast
7 cm strat de nisip



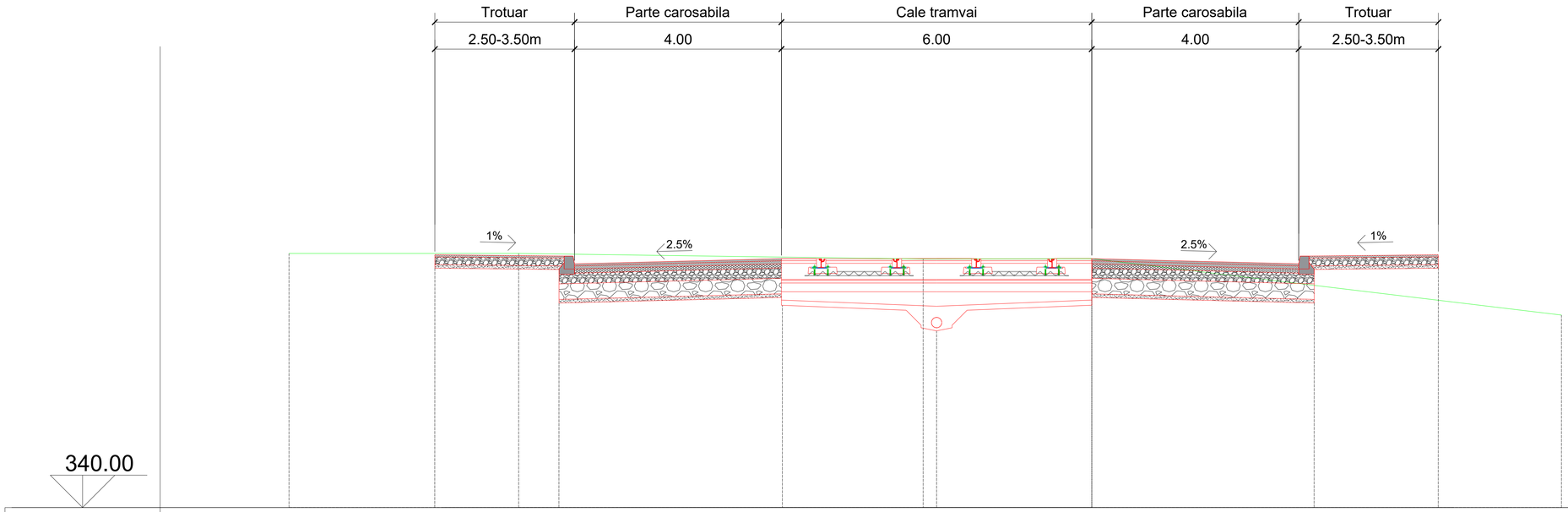
UNIVERSITATEA TEHNICĂ CLUJ-NAPOCA FACULTATEA DE CONSTRUCȚII			PROIECTARE TRAMVAI ȘI METROU ANUL IV / SECȚIA IUDR (3641) AN UNIVERSITAR 2024-2025		PROIECT 1/2025
	NUME	SEMNĂTURĂ	SCARA 1:50	PROFILE TRANSVERSALE	FAZA P.T+D.E
DESENAT	HARAPCIUC BERNARD ALEXANDRU		DATA 15.01.2025		NR. PLANȘĂ
VERIFICAT					


Trotuar
2.50-3.50m
Parte carosabila
4.00
Cale tramvai
6.00
Parte carosabila
4.00
Trotuar
2.50-3.50m
1%
2.5%
2.5%
1%
340.00

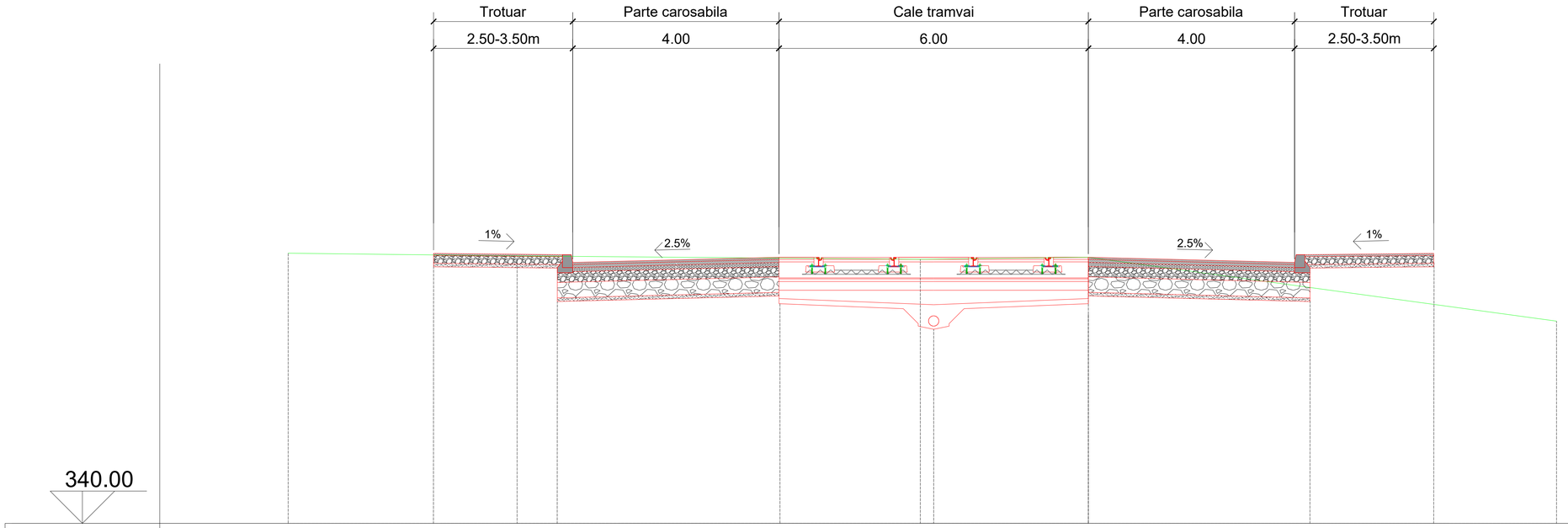
| | | |


Trotuar
2.50-3.50m
Parte carosabila
4.00
Cale tramvai
6.00
Parte carosabila
4.00
Trotuar
2.50-3.50m
1%
2.5%
2.5%
1%
340.00

| | | |

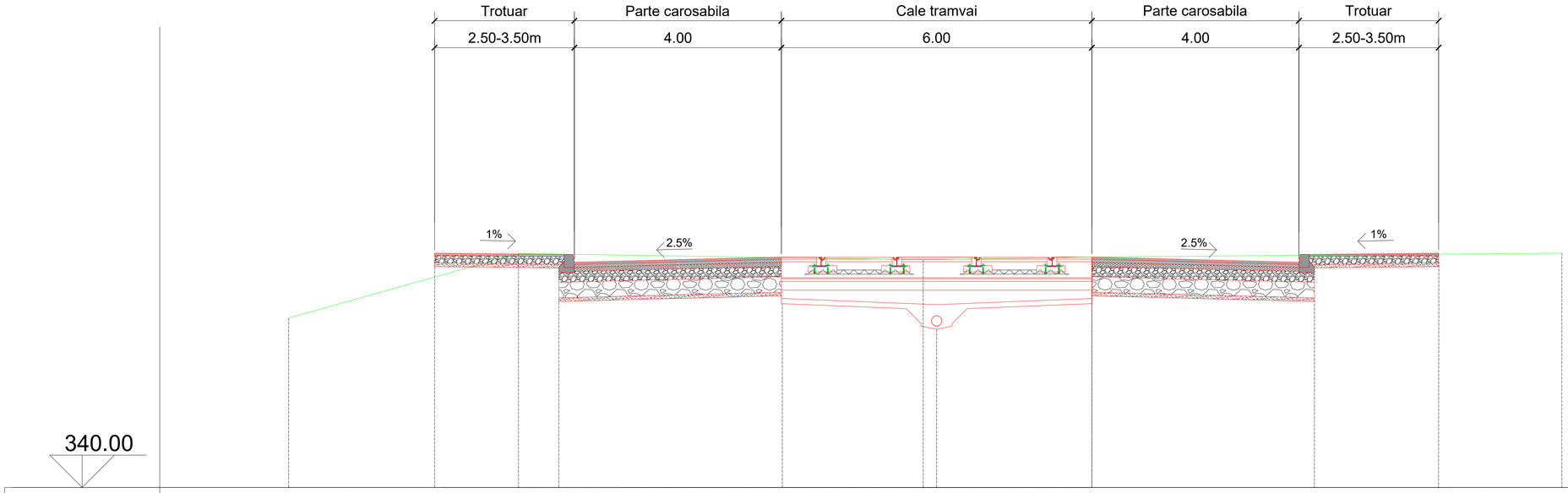

Trotuar
2.50-3.50m
Parte carosabila
4.00
Cale tramvai
6.00
Parte carosabila
4.00
Trotuar
2.50-3.50m
1%
2.5%
2.5%
1%
340.00

| | | |

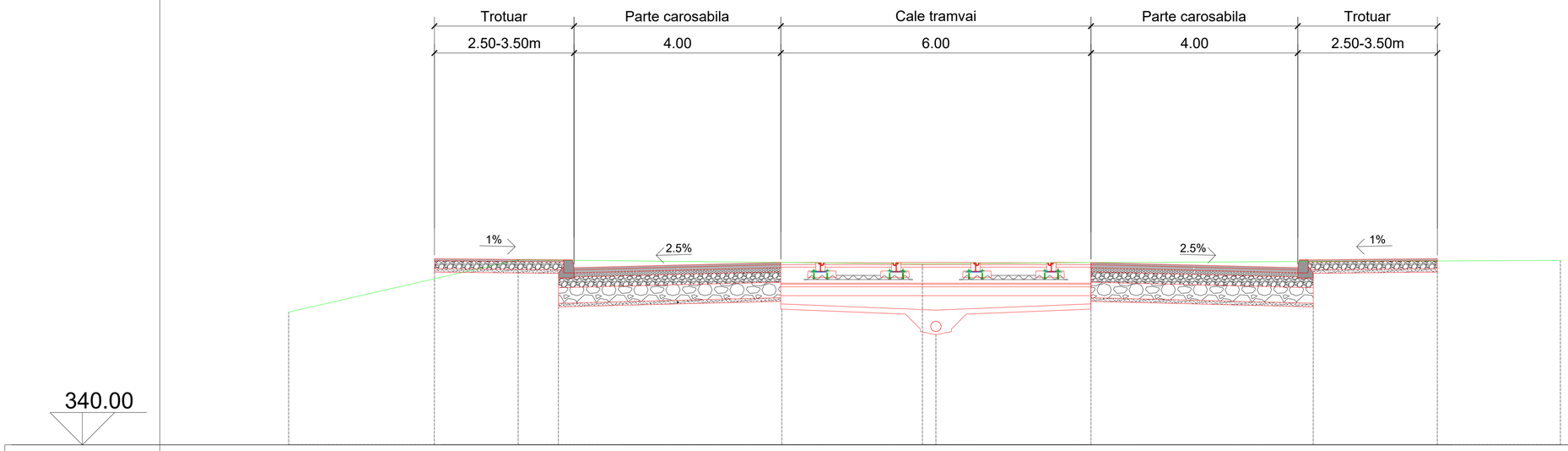


Cotă proiect			345.21	345.21	345.16	345.16	345.16	345.21	345.21	
Cotă teren	345.24		345.18			345.12	345.16			343.92
Distanțe	12.22		8.11			3.32				12.02

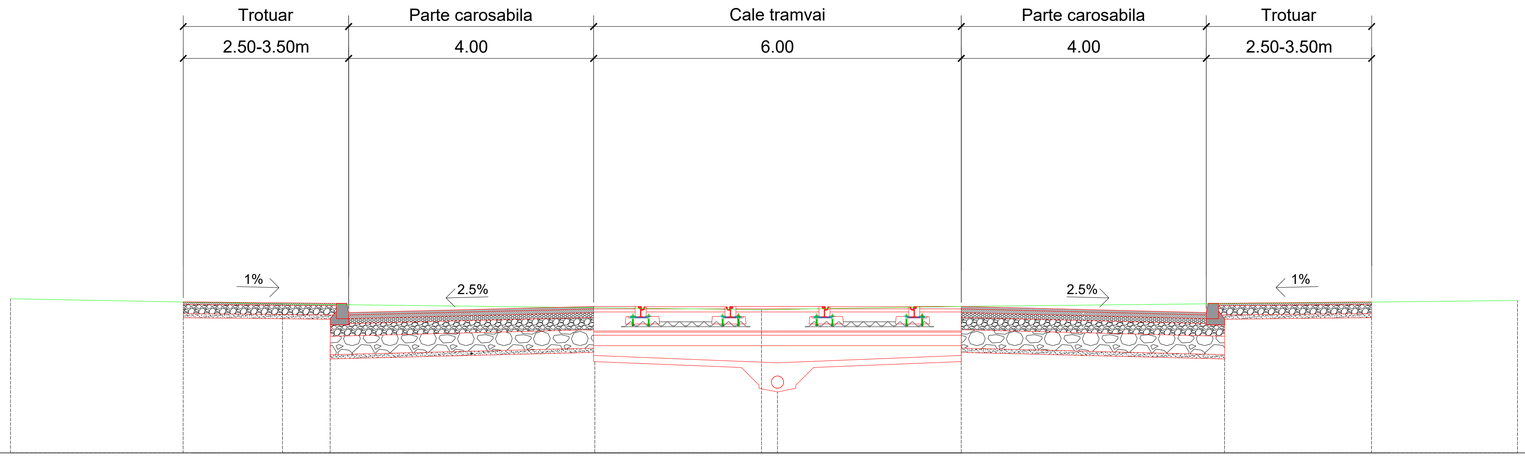
UNIVERSITATEA TEHNICĂ CLUJ-NAPOCA FACULTATEA DE CONSTRUCȚII			PROIECTARE TRAMVAI ȘI METROU ANUL IV / SECȚIA IUDR (3641) AN UNIVERSITAR 2024-2025		PROIECT 1/2025
	NUME	SEMNĂTURĂ	SCARA 1:50	PROFILE TRANSVERSALE	FAZA P.T+D.E
DESENAT	HARAPCIUC BERNARD ALEXANDRU				
VERIFICAT			DATA 15.01.2025		NR. PLANȘĂ


Trotuar
2.50-3.50m
Parte carosabila
4.00
Cale tramvai
6.00
Parte carosabila
4.00
Trotuar
2.50-3.50m
1%
2.5%
2.5%
1%
340.00

| | | |

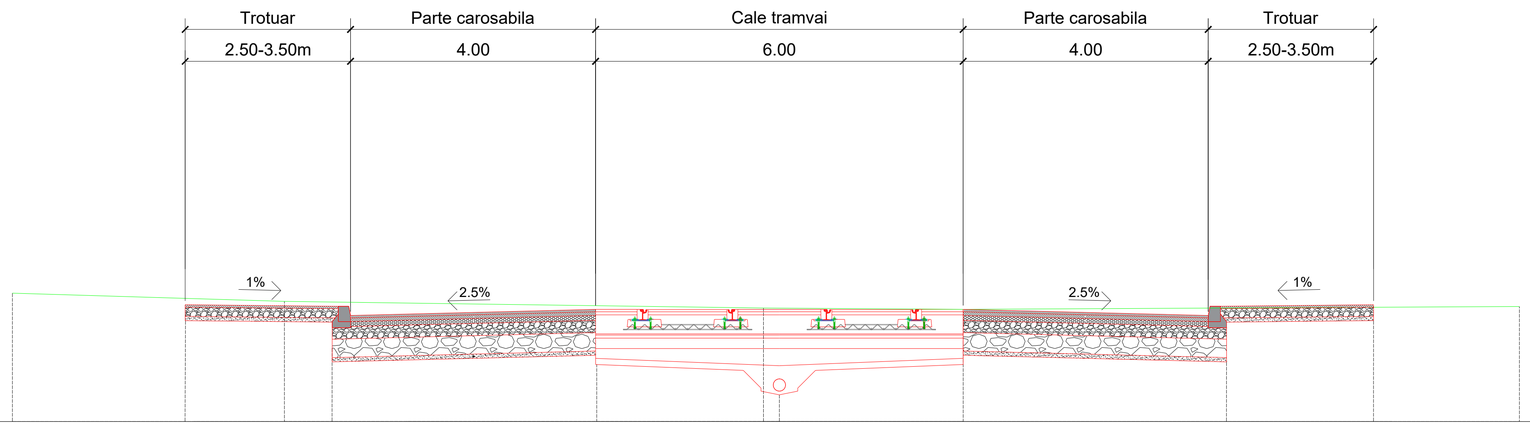

Trotuar
2.50-3.50m
Parte carosabila
4.00
Cale tramvai
6.00
Parte carosabila
4.00
Trotuar
2.50-3.50m
1%
2.5%
2.5%
1%
340.00

| | | |


Trotuar
2.50-3.50m
Parte carosabila
4.00
Cale tramvai
6.00
Parte carosabila
4.00
Trotuar
2.50-3.50m
1%
2.5%
2.5%
1%

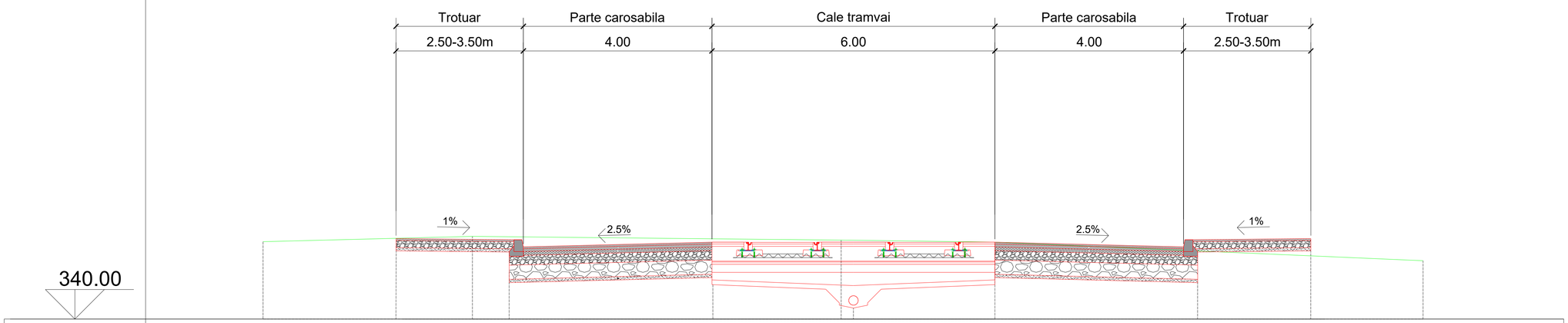
340.00

| | | |

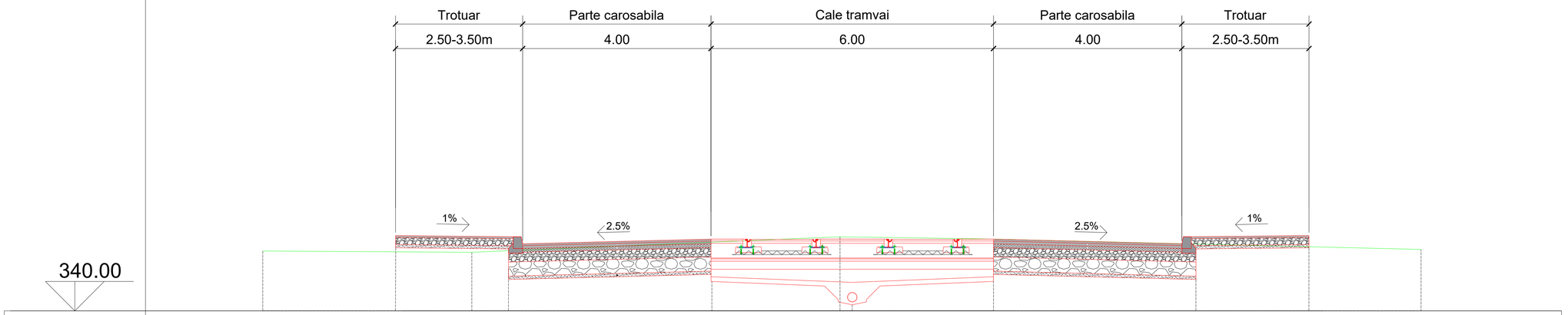

Trotuar
2.50-3.50m
Parte carosabila
4.00
Cale tramvai
6.00
Parte carosabila
4.00
Trotuar
2.50-3.50m
1%
2.5%
2.5%
1%

340.00

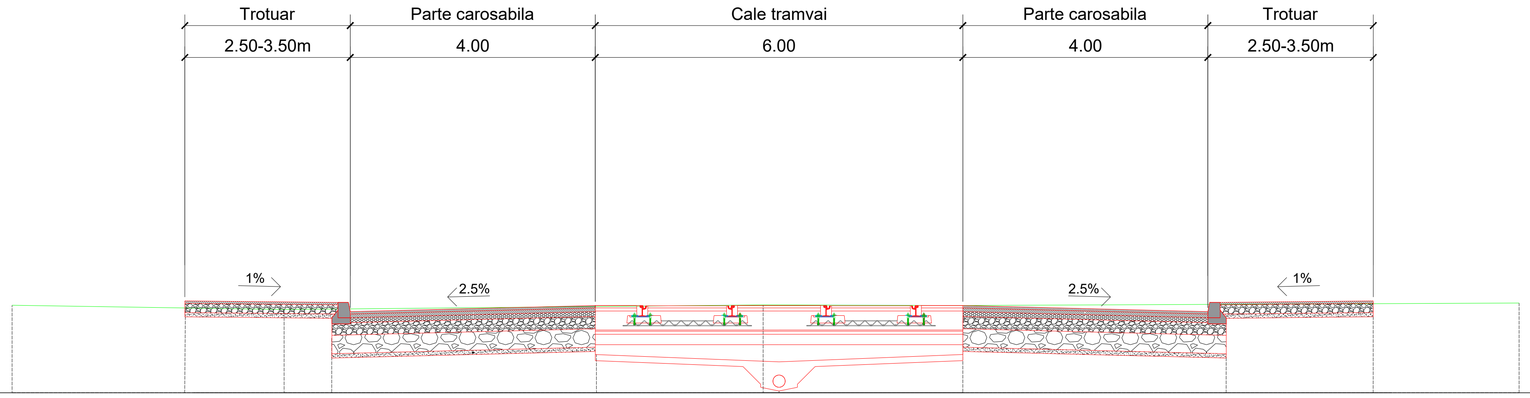
| | | |


Trotuar
2.50-3.50m
Parte carosabila
4.00
Cale tramvai
6.00
Parte carosabila
4.00
Trotuar
2.50-3.50m
1%
2.5%
2.5%
1%
340.00

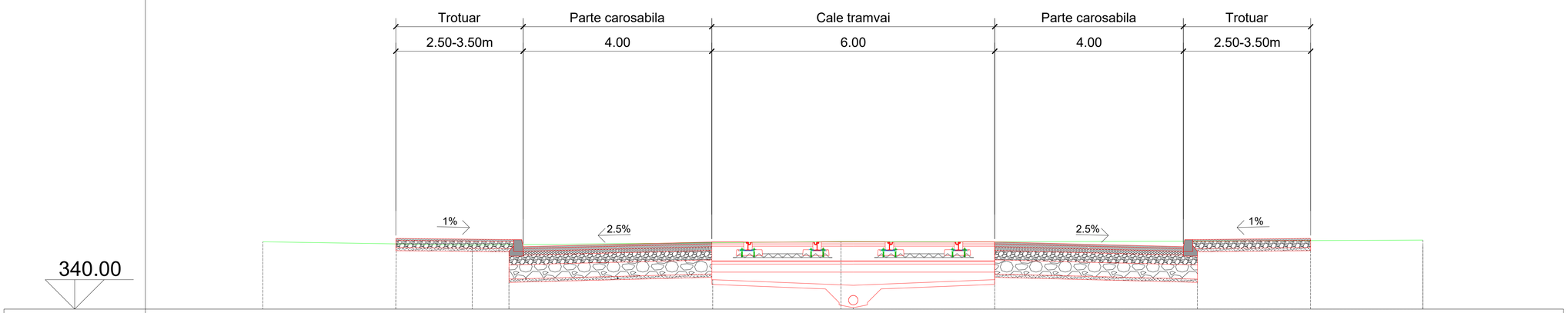
| | | |


Trotuar
2.50-3.50m
Parte carosabila
4.00
Cale tramvai
6.00
Parte carosabila
4.00
Trotuar
2.50-3.50m
1%
2.5%
2.5%
1%
340.00

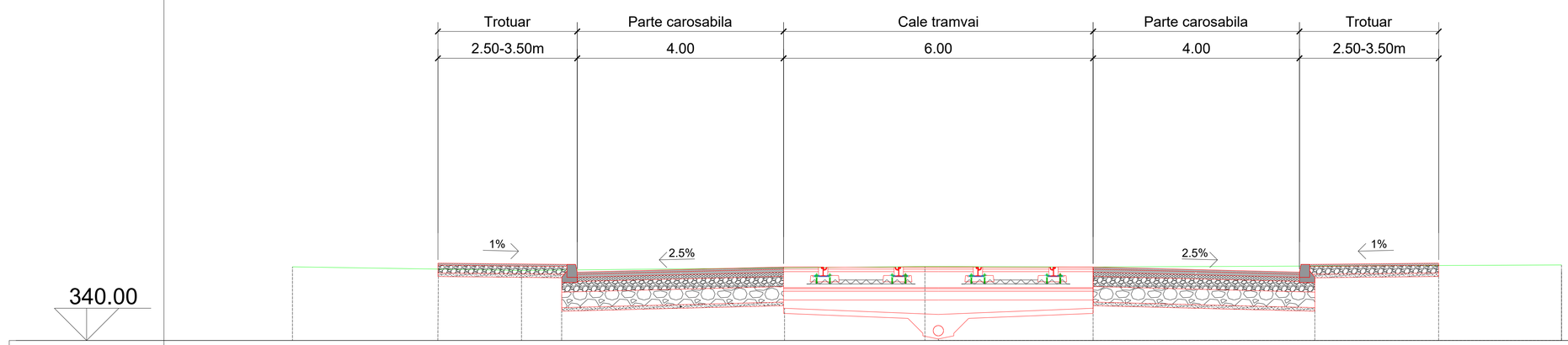
| | | |


Trotuar
2.50-3.50m
Parte carosabila
4.00
Cale tramvai
6.00
Parte carosabila
4.00
Trotuar
2.50-3.50m
1%
2.5%
2.5%
1%
340.00

| | | |


Trotuar
2.50-3.50m
Parte carosabila
4.00
Cale tramvai
6.00
Parte carosabila
4.00
Trotuar
2.50-3.50m
1%
2.5%
2.5%
1%
340.00

| | | |



Cotă proiect	341.08	341.08	341.03	341.03	341.03	341.08	341.08	
Cotă teren	341.28	341.21	341.00	341.03				341.23
Distanțe	12.13	8.13	3.03					12.15

UNIVERSITATEA TEHNICĂ CLUJ-NAPOCA FACULTATEA DE CONSTRUCȚII			PROIECTARE TRAMVAI ȘI METROU ANUL IV / SECȚIA IUDR (3641) AN UNIVERSITAR 2024-2025		PROIECT 1/2025
	NUME	SEMNĂTURĂ	SCARA 1:50	PROFILE TRANSVERSALE	FAZA P.T+D.E
DESENAT	HARAPCIUC BERNARD ALEXANDRU				
VERIFICAT			DATA 15.01.2025		NR. PLANȘĂ